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Convergent Radial-Domain Structure in SPARC Dark Matter Halos Beyond Single Smooth Equilibrium Profiles

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ABSTRACT

Single smooth equilibrium halo profiles leave statistically significant, radially organized residuals in roughly one third of SPARC rotation curves. We introduce a minimum-order radial-domain decomposition: the dark-matter contribution is approximated by a minimum-order sum of tiered cored mass components (Burkert form), with the order selected by the Bayesian Information Criterion. Across 121 galaxies, a second resolved radial mass domain is required in ∼35% of cases — a fraction robust to confining component scales to the measured radii (a constraint that simultaneously collapses all three-domain solutions) and stable across three halo families (Burkert 35%, NFW 37%, free-shape Einasto 32%), though, as for any information-criterion threshold, the precise fraction is threshold-dependent. The fraction is also stable under free stellar mass-to-light ratios (31% versus 35% fixed on the identical sample, with 20 of the 23 decisive systems retained, unchanged when the prior width is doubled): the second domain is not a repackaged global mass-to-light mismatch. The central quantitative result is a bias diagnosis: in all 41 two-domain systems the single-profile scale radius lies between the two domain scales (median factor 2.2) — for these galaxies, single smooth fits are systematically biased summaries, not merely incomplete ones. Convergent evidence supports the detection: galaxies requiring a second domain coincide with galaxies flagged by an algorithmically distinct Gaussian-shell decomposition (odds ratio 38; mass-adjusted 37.5), and the domain crossover radius tracks the independently fitted shell radius ($\rho = 0.73$, permutation $p = 0.004$, surviving half-light-radius control). Forced two-domain fits to single-domain galaxies collapse to degenerate configurations, providing an internal null. Where the structure is strong enough to test, the multi-domain selection reproduces on rotation curves re-derived from the original radio cubes by independent software — decisively ($\Delta\mathrm{BIC} = 121$ for NGC 2403) and at the same crossover radius — consistent with a property of the galaxy, not of the fitting basis. We interpret the second component as a method- and data-robust diagnostic of a second resolved, cored radial mass domain (the term "domain" denoting a radial mass component, with no three-dimensional shape implied, which rotation curves cannot constrain). Of the two components, the empirical signature is carried by the *outer*: it remains consistent with single-domain halo scaling, whereas forced decompositions of single-domain galaxies cannot reproduce that behavior. We make no claim about the inner component's relation to the halo scaling locus — a displacement forcing reproduces — and remain agnostic as to physical origin.

Keywords: Dark matter (353) — Galaxy rotation curves (619) — Galaxy dark matter halos (1880) — Astrostatistics (1882)

1. INTRODUCTION

Rotation curves of late-type galaxies are conventionally decomposed into baryonic contributions plus a single smooth equilibrium dark matter halo, described by cuspy profiles motivated by ΛCDM simulations (NFW; **?**), observationally motivated cored profiles (Burkert; **?**), or intermediate forms (Einasto; **?**). The adequacy of a *single* smooth profile is usually assessed only through global fit quality. Localized departures — coherent residual features at particular radii — have been discussed largely in connection with baryonic features ("Renzo's rule"; **?**), but a systematic, model-agnostic census of whether single smooth halos suffice is lacking.

In a companion analysis (Bibb 2026, Zenodo, https://doi.org/10.5281/zenodo.20263016) we modeled

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localized residual structure in SPARC (?) rotation curves with Gaussian mass shells superposed on a single-Burkert baseline model (hereafter the reference profile), finding BIC-selected shells in roughly half the sample. That decomposition is used here only as a structurally independent comparison basis, not as a source of physical claims. The present paper asks a deliberately minimal, complementary question: *what is the minimum number of smooth, cored radial domains required by each rotation curve?* Because any sufficiently flexible component will absorb residuals — so that inferred "structure" may reflect the chosen basis rather than the galaxy — the credibility of either decomposition rests on validation. We therefore test the answer against the algorithmically distinct shell decomposition, against mass-driven confounds, against an internal forced-fit null, against free stellar mass-to-light ratios, and — for the most decisive systems — against rotation curves of the same galaxies derived independently from the original radio data cubes.

The result is supported by four lines of evidence. First, it is robust: the multi-domain fraction is unchanged when components are forbidden from extrapolating beyond the measured radii, while unstable three-domain solutions are eliminated by the same constraint. Second, it is independent of the reference profile: NFW and Burkert nulls give the same fraction. Third, it is convergent across decompositions: two algorithmically distinct decomposition methods, applied to the same rotation curves, flag the same galaxies and localize structure to consistent radii, and this concordance is not explained by galaxy mass (both methods perturb the same single-profile baseline, so this is convergence of description, not full independence; §?? quantifies the overlap). Fourth, where it can be tested the selection reproduces under independent velocity derivation: on rotation curves re-derived from the original radio cubes by independent software, decisively and at the same radius. A fit improvement on one curve, however constrained the basis, cannot predict the outcome of an independent derivation from a different instrument; that this prediction succeeds is a demanding test of galaxy-intrinsic origin, though one so far established for the strong tail of the population — population-level robustness rests on the mass-to-light and threshold tests above — and is consistent with the structure being a property of the galaxy rather than of the parameterization. We emphasize equally what the analysis does *not* find, and we make no claim about the physical origin or three-dimensional geometry of the detected structure.

2. DATA AND METHODS

2.1. *Sample and baryonic model*

We use the SPARC database (?), restricted to galaxies with quality flag $Q \leq 2$ and at least six rotation-curve points after removing points interior to the first reliably measured radius: 123 galaxies meet these criteria, of which 121 yield converged fits. Baryonic contributions adopt fixed mass-to-light ratios $\Upsilon_{\rm disk} = 0.5$ and $\Upsilon_{\rm bul} = 0.7$ at 3.6 μm with the gas contribution at unity, combined with signed squares so that $V_{\rm bar}^2 = V_{\rm gas}|V_{\rm gas}| + 0.5\, V_{\rm disk}|V_{\rm disk}| + 0.7\, V_{\rm bul}|V_{\rm bul}|$. Velocity uncertainties are floored at $1\,{\rm km\,s^{-1}}$. Points at which the baryonic model exceeds the observed velocity ($V_{\rm obs}^2 \leq V_{\rm bar}^2$) are excluded from the dark-matter fit, matching the companion analysis. The fixed mass-to-light assumption is tested directly by repeating the full selection with $\Upsilon_{\rm disk}$ and $\Upsilon_{\rm bul}$ free (§??).

2.2. *Minimum-order radial-domain decomposition*

The inferred dark-matter contribution is approximated by a minimum-order sum of cored radial mass components of Burkert form,

$$V_{\rm DM}^2(r) = \sum_{i=1}^{N} V_{\rm B}^2(r;\, \rho_{0,i},\, a_i), \tag{1}$$

subject to ordering constraints that enforce a tiered structure and break exchange degeneracies: $\rho_{0,i}$ strictly decreasing and a_i strictly increasing outward. A soft prior ties the total enclosed dark mass at the outermost measured point to the dynamically inferred value; the capped variant below carries no such prior yet returns an indistinguishable multi-domain fraction (35% versus 35.5%; Table ??), providing a prior-free control on the headline number. Fits use bounded nonlinear least squares with 20–24 random restarts; model order is selected by BIC $= \chi^2 + k \ln n_{\rm pts}$ (??), with $k = 2N$ free parameters (two per domain), evaluated for $N = 1$–4. At $N = 1$ the model reduces exactly to a single Burkert halo, providing the null hypothesis within the same machinery.

We adopt cored Burkert components rather than a generic flexible basis (splines, Gaussian processes, or radial basis functions) for four reasons that bear on interpretability. First, each component carries a fixed parameter count (two per domain), so model order is penalized cleanly by the BIC; a spline or Gaussian-process basis does not directly address the scientific question posed here: the minimum number of resolved radial domains. Second, the monotonicity constraints (ρ_0 decreasing, a increasing outward) enforce a physically motivated tiered structure rather than fitting arbitrary residual wiggles. Third, the $N = 1$ case reduces exactly to a single Burkert halo, providing the null within the identical machinery. Fourth, the forced-fit control (§??) requires a basis with an identifiable degenerate limit, which a physically parameterized component possesses and an unconstrained flexible basis does not. The goal is a controlled test of whether more than one resolved radial domain is required, not the best attainable fit.

Two variants are run. In the *unconstrained* variant, scale radii are free. In the *capped* variant, every component is required to satisfy $a_i \in [r_{\min}, r_{\max}]$, the measured radial range, so that no component's characteristic scale lies where there are no data. The cap is not a technical variant but a *stability discriminator*: structure that survives it is resolved by the data, while structure that collapses is identified as extrapolation artifact. It thereby provides an internal falsification mechanism for each candidate domain.

2.3. *Comparison methods*

Gaussian-shell decomposition. The companion method models localized residuals as Gaussian mass shells of center r_s, width σ, and mass M_s on a single Burkert reference profile, with shell count BIC-selected and a width constraint $\sigma/r \le 0.4$ that defines locality. The preferred fractional width is cap-invariant: the median $\sigma/r \simeq 0.22$ is unchanged as the allowed ceiling is raised from 0.4 to 1.0, with only 4% of shells near the ceiling at a cap of 1.0. The shell and domain decompositions share data but use structurally dissimilar basis functions — a localized annular perturbation versus a global cored profile — and are fit independently.

Reference-profile generalization. The minimum-order analysis of §?? is repeated with NFW and Einasto in place of Burkert. For Einasto we run two variants: a fixed-shape null ($\alpha = 0.16$) and a free-shape variant in which α is an additional shared parameter ($k = 2N + 1$, with the extra parameter charged to the BIC), bounded to $\alpha \in [0.05, 0.50]$, spanning and exceeding the range observed in simulated and fitted halos.

3. RESULTS

3.1. *A robust two-domain population*

Table ?? summarizes the BIC-selected model order. Unconstrained: 78/38/5 galaxies select $N = 1/2/3$ (64/31/4%). Capped: 79/41/0 (one galaxy selects $N = 4$). The multi-domain fraction is essentially unchanged, 36% versus 35%, and 95% (41/43) of multi-domain galaxies retain $N \ge 2$ under the cap. By contrast, *all five* three-domain solutions collapse: in every case the third component was an inner spike below the first measured point or an outer ramp beyond the last, i.e. extrapolation rather than measurement. The data support at most two resolved radial domains.

Table 1. Multi-domain fraction ($N \ge 2$) under alternative model-selection criteria, re-scored from the existing fits. The population persists across every standard threshold; the fiducial analysis uses the 2-unit BIC margin.

Selection rule	$N \ge 2$	Fraction
BIC, standard	43	35.5%
BIC, 2-unit tie band	40	33.1%
BIC, $\Delta > 6$	36	29.8%
BIC, $\Delta > 10$ (very strong)	32	26.4%
AIC	48	39.7%

The ΔBIC margins are strongly asymmetric. Where two domains win, they win decisively: median $\Delta\text{BIC} = +43.3$ over the single profile, with 23/42 cases exceeding $+30$ and a maximum of $+1115$. Where the single profile wins, it never wins decisively: the median losing margin is $+4.8$, 81% of losses are within $\Delta\text{BIC} < 6$, and *no* galaxy prefers a single Burkert by $\Delta\text{BIC} > 10$. The single-domain population is therefore adequate rather than required, and the quoted 35% should be read as threshold-dependent in the usual information-criterion sense.

The multi-domain fraction is nonetheless not an artifact of the particular BIC threshold. Re-scoring the one- versus two-domain decision from the existing fits (no re-fitting) across a range of thresholds and under the AIC, the fraction preferring a second radial domain varies smoothly and modestly across every standard criterion (Table ??): from 26% under a stringent $\Delta\text{BIC} > 10$ cut — the ? "very strong" threshold — to 40% under the more permissive AIC, bracketing the fiducial 35%. The population never collapses. The decision-margin asymmetry described above is likewise threshold-independent: the decisive two-domain wins and uniformly marginal single-domain preferences persist under every criterion in Table ??. A basis whose two-domain preference reflected mere fitting flexibility would produce symmetric, marginal preferences; the observed asymmetry does not.

A direct consequence concerns what a single profile reports when two domains are present. In all 41 capped two-domain systems, the single-Burkert scale radius lies between the inner and outer domain scales, with a median $a_{\text{Burkert}}/a_{\text{in}} = 2.19$, indicating that a single smooth reference profile compresses the two-domain structure into a biased intermediate scale rather than recovering the inner domain plus a residual correction. Smooth fits can therefore not only conceal radial structure but sys-

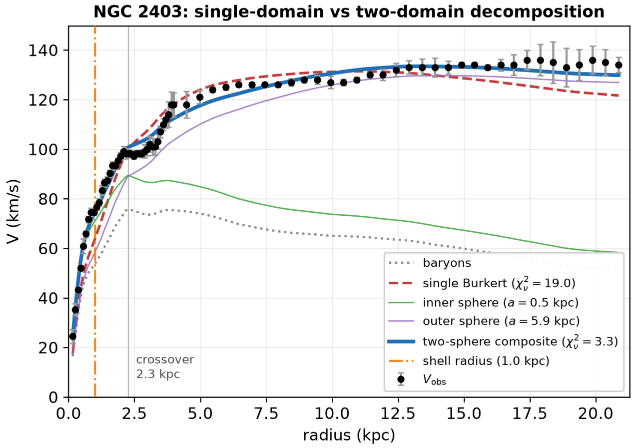

Figure 1. Representative two-domain decomposition (NGC 2403, SPARC). The observed rotation curve (points) and baryonic contribution (dotted) are shown with the best single-Burkert fit (dashed) and the BIC-selected two-sphere composite (thick solid); the inner and outer cored components are shown individually (thin solid). The single profile cannot reproduce both the inner rise and the outer profile simultaneously, whereas the two-domain composite does (χ^2_ν improving from $\simeq 19$ to $\simeq 3$). The domain crossover (vertical line, $\simeq 2.3$ kpc) lies close to the radius of the independently fitted Gaussian shell (marked, $\simeq 1.0$ kpc) for this galaxy, previewing the population-level concordance of Figure **??**. This is also one of the systems for which the selection reproduces on a rotation curve re-derived from the original radio cubes by independent software (§**??**).

Table 2. BIC-selected number of radial domains

Variant	N=1	N=2	N=3	N≥2 frac.
Unconstrained	78	38	5	36%
Capped ($a \in [r_{\rm min}, r_{\rm max}]$)	79	41	0^a	35%
NFW reference profile (unconstrained)	76	36	9	37%
Einasto reference profile^b	56	47	18	54%

[a] One galaxy selects $N = 4$.

[b] Fixed $\alpha = 0.16$. The inflated fraction is a fixed-shape artifact: freeing α (range $[0.05, 0.50]$, with the extra parameter BIC-charged; one galaxy, UGC 06923, fails free-α convergence and is excluded) returns 32% (§**??**), consistent with Burkert and NFW. The fitted α rails at the upper bound of 0.50 in the majority of galaxies (89/120), indicating the shape freedom is spent absorbing inner structure with one steep component rather than adding a domain — the mechanism of the fixed-shape inflation. Even against a maximally flexible single-Einasto null (α free to the same 0.50 bound) the two-domain fraction remains 28%, below the Burkert fraction, so the two-domain detection is conservative against an over-flexible null.

tematically bias inferred core scales, by a factor of $\sim$2 in this sample. Figure **??** shows a representative example.

3.2. *Reference-profile independence and comparison to single profiles*

With an NFW reference profile the multi-domain fraction is 37%, statistically indistinguishable from the Burkert 35–36%: the residual structure is not an artifact of fitting a cored profile to cuspy halos or vice versa. The fixed-shape Einasto null returns 54%, but this is a rigidity artifact: freeing the Einasto shape parameter α returns 32% (Table **??**, note b). The two-domain fraction is therefore stable at 32–37% across all three halo families once each is given a fair shape, with the fixed-α Einasto 54% identified as the outlier rather than evidence of extra structure.

For the capped two-domain galaxies, the composite outperforms every single-profile alternative on BIC, including the parameter penalty. Scoring wins/losses/ties with a 2-unit BIC margin — below which the models are treated as statistically indistinguishable (the **?**, convention) — the composite records 36/0/5 against a single Burkert, 27/7/7 against a single NFW, and 32/7/2 against a single Einasto: it never loses to a single Burkert and loses to NFW or Einasto in at most seven cases. (For single-domain galaxies the composite is by construction identical to a single Burkert, so that comparison is degenerate; it is reported as non-applicable rather than scored.) The inferred multi-domain population is not eliminated by replacing one smooth halo family with another.

3.3. *Galaxy-level concordance with the shell method, controlling for mass*

The central result is cross-method agreement. Of the 42 capped multi-domain galaxies, 39 are independently shell-bearing; of the 79 single-domain galaxies, 59 are shell-free (Fisher exact $p < 10^{-4}$; odds ratio 38; overall agreement 81%).

Because both structure indicators correlate with galaxy mass, we test whether the concordance is a mass artifact. In logistic regression for multi-domain selection with $\log M_\star$ as the base predictor, shell-bearing status adds likelihood-ratio $\chi^2 = 49.7$ ($p < 10^{-4}$) — an order of magnitude beyond any other covariate — while T-type, effective surface brightness, $R_{\rm HI}$, gas fraction, and bulge presence (LR $\chi^2 = 0.01$, $p = 0.93$) are entirely absorbed by mass, and $V_{\rm flat}$ contributes only weakly ($p = 0.013$). The bulge null additionally disarms the concern that the second component merely compensates for bulge–disk mass-to-light errors: if it did, bulge presence would predict multi-domain selection at fixed mass, and it does not. Stratifying into $\log M_\star$ tertiles, the shell–domain association holds within every bin (Fisher $p = 0.011$, $< 10^{-4}$, $< 10^{-4}$), and the Cochran–Mantel–Haenszel mass-adjusted odds ratio is 37.5 — indistinguishable from the raw 38. At fixed mass, shell-bearing galaxies are $\sim$37 times more likely to require a second radial domain.

3.4. *Radial-level concordance*

Co-occurrence in the same galaxies could still mean the two methods describe different features. We therefore compare *where* they place structure. For each capped two-domain galaxy we compute the crossover radius $r_\times$ at which the outer component's velocity contribution overtakes the inner, and compare it to the independently fitted shell radius (nearest shell; $n = 28$ galaxies host both).

The two radii are strongly correlated: Spearman $\rho = 0.73$ ($p < 10^{-4}$), with a median fractional offset $|r_\times - r_s|/r_{\rm max} = 0.188$ against 0.387 for permuted shell assignments (permutation $p = 0.0038$). Both quantities partly track the optical half-light radius ($\rho = 0.57$ for $r_\times$ vs. $R_{\rm eff}$), so we control for it: residualizing both radii on $R_{\rm eff}$, the shell–crossover correlation persists at partial $\rho = 0.62$ ($p = 4\times10^{-4}$). The two methods localize structure to common radii for reasons beyond a shared preference for the mid-disk. Notably, the capped fits show *stronger* radial concordance than the unconstrained fits ($\rho = 0.73$ vs. 0.68; permutation $p = 0.004$ vs. 0.047), as expected if the cap removes extrapolated components that scatter the crossover radius. The galaxy-by-galaxy correspondence is shown in Figure **??**.

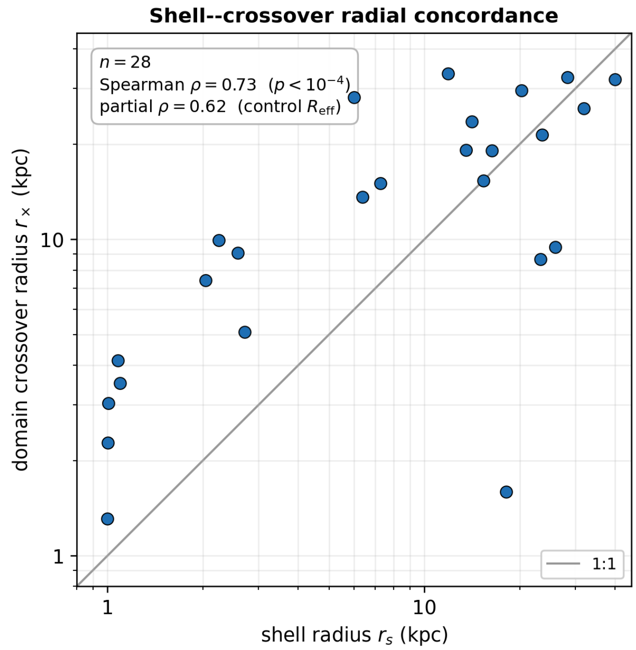

Figure 2. Radial concordance between the two decompositions. For the $n = 28$ galaxies hosting both a BIC-selected second domain and an independently fitted shell, the domain crossover radius $r_\times$ is plotted against the nearest shell radius r_s (logarithmic axes); the solid line is $r_\times = r_s$. The two radii are strongly correlated (Spearman $\rho = 0.73$), and the correlation survives control for the optical half-light radius (partial $\rho = 0.62$), so the two methods localize structure to common radii for reasons beyond a shared preference for the mid-disk.

3.5. *An internal null: forced two-domain fits*

As an internal control, we force $N = 2$ fits on the 79 galaxies whose BIC-selected solution is single-domain. In 71/79 the added component yields negligible likelihood improvement ($\Delta\chi^2 < 2$); only 8/79 improve genuinely but fail the BIC penalty. The forced components collapse toward degenerate configurations: median $a_{\rm out}/a_{\rm in} = 1.16$, versus 9.09 in selected two-domain systems ($p < 10^{-4}$), with the added component's density suppressed by $\sim$4 dex. Moreover, the inner–outer scale coupling present in selected systems survives control for galaxy size (partial $\rho = 0.56$, $p = 10^{-4}$), whereas in forced fits it vanishes (partial $\rho = 0.13$, $p = 0.25$). Forced decompositions therefore converge toward a characteristic degenerate configuration, distinct from the selected multi-domain population.

A further property of the forced fits sharpens this null into a positive discriminant. Placing both the forced and the genuinely selected components against the single-domain halo ρ_0–scale relation — built locally in $V_{\rm flat}$ to remove the mass-regime dependence — the two component tiers behave asymmetrically under forcing. The *inner* components of forced and selected fits are statistically indistinguishable in their displacement below the relation (Mann–Whitney $p = 0.10$): forcing any second component drives an inner term below the relation, so inner displacement alone is *not* diagnostic of genuine structure, and we make no claim on it. The *outer* components, by contrast, separate cleanly: in selected two-domain systems the outer component remains comparatively close to the single-domain locus (median -0.95σ), whereas in forced decompositions — excluding degenerate fits that rail against parameter bounds — it is driven far below it (median -3.44σ; Mann–Whitney $p = 2 \times 10^{-4}$, KS $p = 2 \times 10^{-4}$). These offsets are computed from the capped fits, which carry no enclosed-mass prior, so the asymmetry cannot be inherited from the soft prior of §**??**. Were the decomposition mere fitting freedom, both tiers would respond to forcing alike; instead one tier survives the forcing and the other does not. That asymmetry is itself evidence that the outer component of a selected system is the portion of the decomposition that survives every internal validation applied here, rather than an artifact of adding parameters.

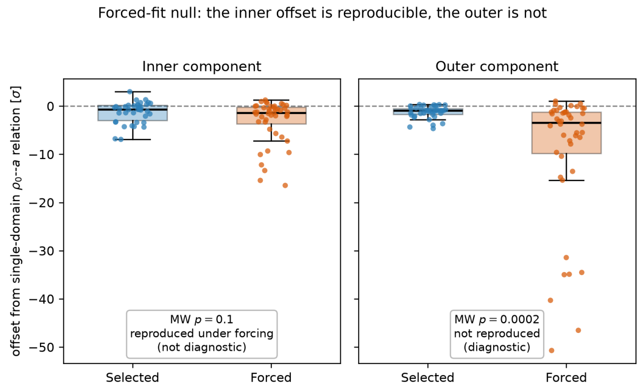

Figure 3. The forced-fit null and its asymmetry. For single-domain galaxies, a second component is *forced* and compared with the components of genuinely BIC-selected two–domain systems. The inner components are statistically indistinguishable in their displacement below the single-domain ρ_0–scale relation (Mann–Whitney $p = 0.10$): forcing reproduces the inner offset, so it is not diagnostic. The outer components separate cleanly — selected systems remain near the locus (median -0.95σ) while forced fits are driven far below it (median -3.44σ; $p = 2 \times 10^{-4}$). One tier survives forcing and the other does not, isolating the outer component as the one that survives every internal validation applied here.

Figure ?? contrasts the selected and forced populations directly.

3.6. *What does not predict the second domain*

Three deliberately reported negative results bound the interpretation.

No privileged scale ratio. The capped outer/inner scale ratio has median $a_{\rm out}/a_{\rm in} \simeq 9$, but it does not correlate with the dimensionless structural ratios $R_{\rm HI}/R_{\rm d}$ or $r_{\rm max}/R_{\rm d}$ ($|\rho| \leq 0.13$, $p > 0.4$); it correlates only with absolute size and mass. The absolute scales track the measured extent ($a_{\rm out}$ vs. $r_{\rm max}$: $\rho = 0.80$). The decomposition scales with the galaxy; it does not identify a special radius ratio.

No common surface-density manifold. Pooled components follow a ρ_0–a anticorrelation of slope -1.29, superficially resembling the constant surface-density relation (?), but inner and outer components occupy significantly different surface-density levels ($\log \rho_0 a = 8.59$ vs. 7.98; $p < 0.001$). The two tiers are not a single scaling law sampled twice.

No gas-fraction signal at fixed mass. Within mass tertiles, multi-domain and single-domain galaxies have indistinguishable gas fractions ($p = 0.13$–0.96).

3.7. *Robustness to mass-to-light freedom*

The fixed mass-to-light ratios of §?? are a potential structured confound rather than a calibration detail: a galaxy whose true $\Upsilon_{\rm disk}$ departs from the population value carries a smooth, disk-scale velocity residual that a second cored component could absorb, and the forced-fit null of §?? does not close this channel, because such a residual is a genuine feature of the data rather than fitting flexibility. We therefore repeat the minimum-order selection with $\Upsilon_{\rm disk}$ and $\Upsilon_{\rm bul}$ free under lognormal priors of width 0.1 dex about the nominal values, following the now-standard SPARC treatment (?), with the additional parameters charged identically to both model orders so that the BIC comparison is unaffected by the added freedom.

The selection survives, and the comparison chains directly to the fiducial numbers. Re-run at fixed Υ on the identical 121-galaxy sample, the selection engine reproduces the fiducial multi-domain fraction (43/121, 35.5%; cf. Table ??). With Υ free the fraction is 38/121 (31.4%), and 20 of the fiducial analysis's 23 decisively multi-domain systems (ΔBIC > 30; §??) retain their selection. The result is not an artifact of the adopted prior width: doubling it to 0.2 dex leaves the decisive retention unchanged (20/23) and moves only the marginal population (fraction 28.1%) — the prior sensitivity is confined to systems whose selection was marginal at fixed Υ. The fitted disk mass-to-light ratios of the surviving multi-domain systems cluster at the nominal value (median $\Upsilon_{\rm disk} = 0.48$, 10–90% range 0.33–0.61): the second domain does not persist by displacing the stellar mass scale. Two attenuations accompany this and should be stated plainly. Selection margins weaken — the median ΔBIC among retained multi-domain systems falls from 65 to 23 — so mass-to-light freedom absorbs part, but not the core, of the residual structure. And membership shifts at the margins (ten exits, six entries relative to the fiducial 42); notably NGC 3521, the honest non-reproduction of §??, exits under free Υ at both prior widths: two independent probes agree that its domain signal is marginal. Finally, the scope of this test should be exact: a single free $\Upsilon_{\rm disk}$ per galaxy closes the *global* mass-to-light channel, but a radial Υ gradient produces a structured residual that no global rescaling, free or fixed, can flatten; that channel is addressed in §??.

3.8. *Reproduction on independently derived rotation curves*

The preceding tests establish that the second domain is independent of the fitting basis. They do not, by themselves, exclude the possibility that the structure originates in the specific rotation curves published by SPARC — for example in a localized systematic of one velocity-field extraction. We therefore test a demanding validation: whether the multi-domain selection reproduces when the rotation curve is re-derived from the original radio data cubes by independent software, holding the baryonic model fixed so that only the measured velocities change.

For the dwarf control galaxies DDO 154 and DDO 168, asymmetric-drift-corrected circular-velocity curves derived by an independent group from the same survey cubes (3D forward-modeling; **?**) reproduce the single-domain selection ($N = 1$) in both cases. On the multi-domain side, we re-derived curves for NGC 2403 and NGC 3198 directly from public THINGS-class cubes using independent tilted-ring forward-modeling software (^{3D}Barolo; **?**), with geometry fixed to literature tilted-ring solutions. Both reproduce the multi-domain selection decisively: NGC 2403 selects $N = 2$ at $\Delta\mathrm{BIC} = 121$ with the domain crossover at 1.9 kpc (SPARC-based value $\sim$ 2.3 kpc), and NGC 3198 selects $N = 2$ at $\Delta\mathrm{BIC} = 55$. For NGC 2403 the independent curve also derives from different observations than those underlying the SPARC curve, strengthening the test. A third multi-domain galaxy, NGC 3741, reproduces at lower significance on an independent VLA-ANGST extraction (different telescope from the WSRT-based SPARC curve; **?**): the multi-domain geometry recovers, though the modest signal-to-noise of this dwarf limits the ΔBIC margin.

Two further multi-domain galaxies, NGC 2841 and NGC 3521, are strongly warped, so a blind fixed-geometry extraction cannot test them; we instead use published warp-aware tilted-ring rotation curves (DiskFit; Sellwood & Spekkens, private communication) derived from the same THINGS cubes. **NGC 2841 reproduces**: the warp-aware curve selects $N = 2$ at $\Delta\mathrm{BIC} = 11$. Its earlier failure on a blind extraction is attributable to the warp suppressing the feature, which the warp-aware curve recovers. **NGC 3521 does not reproduce**: its warp-aware curve prefers a single domain ($\Delta\mathrm{BIC} = -7$ against two), an honest non-reproduction that we report as such. An injection–recovery power control on the warp-aware curves confirms both verdicts are informative: the two-domain feature these galaxies would need to show perturbs the curve by $\sim$20–50 $\mathrm{km\,s^{-1}}$, well above the $\sim$13–19 $\mathrm{km\,s^{-1}}$ amplitude their sampling and uncertainties can detect, so neither is power-limited.

A single galaxy, NGC 1003, is genuinely instrument-limited and we report it as such with a quantified justification: its independent (HALOGAS) extraction yields only 14 usable rings, and an injection–recovery power control shows that the two-domain feature it would carry (peak deviation $\sim$3.3 $\mathrm{km\,s^{-1}}$ over a single-domain fit) lies below the $\sim$7 $\mathrm{km\,s^{-1}}$ amplitude its sampling and noise can detect. Its non-reproduction is therefore uninformative — the curve is incapable of resolving the feature — rather than a negative result.

The adjudicated tally is thus four reproductions (NGC 2403, NGC 3198, NGC 2841, NGC 3741) spanning three classes of independence (same cubes/different software, different observations, different telescope), one honest non-reproduction (NGC 3521), one power-limited indeterminate case (NGC 1003, quantified), and two single-domain controls reproduced (DDO 154/168). Reproduction holds in four of the five adjudicable galaxies; the one failure and the one exclusion are reported explicitly rather than absorbed into an undifferentiated "instrument-limited" category.

We frame this as proof of principle rather than population-level validation: the adjudicable sample is small (five multi-domain systems plus two single-domain controls), limited by the subset of SPARC galaxies whose rotation curves can be re-derived from public cubes at sufficient quality. It establishes that the selection *can* reproduce under genuinely independent velocity derivation, not the rate at which it does across the full sample; the latter awaits wider cube availability.

3.9. *Relation to the shell decomposition: absorption by the domain model*

The galaxy- and radius-level concordance of §**??**–**??** shows that the shell and domain methods flag the same galaxies at the same radii. A sharper question is whether the localized shells carry information *beyond* the broad two-domain structure, or whether they are largely a localized re-parameterization of it. We test this directly by freezing the BIC-selected sphere-domain model of each galaxy and asking whether Gaussian shells, fit to the residual of that frozen model using the identical shell machinery and BIC penalty of the companion analysis, survive.

They mostly do not. Among the capped two-domain galaxies, 21 of 28 shells that are selected against a single-Burkert reference profile are absorbed once the second domain is modeled — they no longer improve the fit beyond the BIC penalty. Population-wide, shell detections fall from 33 to 11 once the domain model is in place.

This reduction is not the trivial consequence of adding a more flexible model. The two-domain model is fit first and then held fixed; the shells are fit to its residual under the identical BIC penalty used in the companion analysis. A shell that merely improved the fit through added freedom would still be BIC-selected against the residual. That most shells instead fail to survive indicates that the localized feature they encoded was already captured by the broad two-domain structure — the two bases describing the same feature — rather than that a more flexible model has absorbed an independent one. Only three galaxies retain a shell at $\Delta\mathrm{BIC} > 10$ against the frozen two-domain model; in these, the surviving shell sits well away from the domain crossover, consistent with a genuinely localized feature distinct from the broad structure. The implication is twofold: most localized "shells" in the companion analysis are the broad two-domain structure expressed in a localized basis — which is why a localized basis drew the objection that it merely improves the fit — and the domain decomposition is the more economical description, with a small residual minority of genuinely localized features isolated for separate study.

4. DISCUSSION

Table **??** summarizes the alternative explanations considered for the multi-domain population, the test applied to each, and the outcome; the final row lists the channels this analysis does not exclude.

The evidential chain is: (1) a single smooth equilibrium profile — whether NFW or Burkert — leaves BIC-significant residual structure in ∼35% of SPARC galaxies; (2) the structure is limited to two resolvable radial domains, with three-domain solutions exposed as extrapolation by the data-range constraint; (3) an unrelated localized-shell decomposition flags the same galaxies, and the association is untouched by mass control; (4) the two methods place structure at consistent radii, beyond both a permutation null and a generic mid-disk scale; (5) no other measured galaxy property predicts the second domain; (6) where it can be adjudicated, the selection reproduces on rotation curves re-derived from the original radio cubes by independent software in four of five cases (including warp-aware curves), with the single non-reproduction and one power-limited case reported explicitly; and (7) when the broad two-domain model is fit first, it absorbs the large majority of the localized shells, identifying them as the same structure in a localized basis rather than independent features.

We emphasize what is and is not being claimed. The present analysis does not imply that galaxies physically contain discrete Burkert subhalos, and it does not claim decomposition *uniqueness*. It claims decomposition *convergence*: structurally independent decomposition bases repeatedly recover the same subset of galaxies as inconsistent with a single smooth equilibrium profile. We accordingly interpret the second component as a *diagnostic of a second radial mass domain*, not as a literal nested halo: the negative results of §**??** argue specifically against reading the components as physical objects with preferred scale ratios or a shared structural relation. A preliminary non-parametric density inversion on a demonstration subset is consistent with this picture — the freely fit $\rho(r)$ shows low-order curvature rather than fragmenting into features at arbitrary radii — though we defer its quantitative development. If multiple mathematically distinct representations ultimately recover comparable transition radii, the evidence would favor interpreting the radial domains as properties of the data rather than of any individual parameterization.

A second boundary on interpretation is geometric and absolute. A rotation curve constrains only the enclosed-mass profile $M(<r)$ through $V^2(r) = GM(<r)/r$; it carries no information about how the mass producing a given $M(<r)$ is arranged in angle or in the third dimension. A second radial domain, a spherical shell, a planar ring, a stream, a warp-induced redistribution, and a localized deposit are indistinguishable in $V(r)$ if they enclose the same mass within each radius. We therefore make no claim about the three-dimensional geometry of the detected structure: the components of our decomposition are radial mass-profile features, and the labels "domain" and "shell" denote parameterizations, not shapes. Discriminating a centrally concentrated from a localized geometry requires observables that carry angular information — azimuthal structure in the velocity field or in resolved tracers — and is deferred to analyses of that kind. The present claim is confined to what rotation curves can support: a reproducible, basis-independent feature in the enclosed-mass profile.

The forced-fit control of §**??** makes the basis-flexibility question empirical. Forced two-domain fits to single-domain galaxies collapse toward degenerate same-scale, vanishing-density configurations with negligible likelihood gain; BIC-selected two-domain systems instead exhibit separated inner and outer scales with residual scale coupling at fixed galaxy size. The detected structure is therefore not a generic consequence of basis-function flexibility: the basis has a recognizable signature for "nothing here," and the selected population does not show it.

Nor should single-domain galaxies be read as negative detections. The margin asymmetry of §**??** — decisive multi-domain wins against uniformly marginal single-profile wins — indicates heterogeneous structural complexity across the population: some systems clearly require additional radial structure, while many are merely *adequately* described by a single profile rather than positively inconsistent with further structure. Whether the detected structure's origin is baryonic (e.g. features propagated from the gas disk; **?**), kinematic, or a gen-

Table 3. Alternative explanations and controls

Alternative explanation	Test	Outcome	
olation beyond measured radii	Cap all scales to $[r_{\min}, r_{\max}]$	Fraction unchanged (35%); all $N \geq 3$ excluded	§?
nce-profile choice	NFW and Einasto refits	32–37% across families	§?
shape rigidity	Shared free $\alpha \in [0.05, 0.50]$	32%; α rails at 0.50	Ta
ed-mass prior	Capped variant carries no prior	35% versus 35.5%	§?
stellar mass-to-light	Free Υ, lognormal prior; doubled width	31%; 20/23 decisive systems retained	§?
c fitting flexibility	Forced N=2 on single-domain galaxies	Degenerate collapse; outer asymmetry $p = 2 \times 10^{-4}$	§?
old choice	Selection-criterion ladder	26–40% across criteria; 23/121 decisive at $\Delta\mathrm{BIC} > 30$	Ta
hoice[a]	Localized Gaussian-shell basis	Same galaxies (odds ratio 38); same radii ($\rho = 0.73$)	§?
y-extraction systematics	Curves re-derived from radio cubes	Reproduced for the decisive tail; marginal population untested	§?
nd size scaling	Mass-matched and partial-correlation controls	Concordance retained	§?
Υ gradients; bar/spiral streaming	Not tested here	Not excluded; deferred	§?

ell and domain bases perturb a common single-profile baseline (§??); this row records convergence of description, not independence of method.

uine property of the dark matter distribution is not determined here, and we decline to speculate. The single external fact available is that, at fixed mass, the second domain co-occurs with localized shell structure at odds ratio ∼37.

We address directly the objection that a constrained multi-component fit might simply be an elaborate means of minimizing χ^2, reproducible by any sufficiently flexible function. Taken together, the bounded model order (the capped selection settles at 65/35/0 rather than consuming the sample toward ever-higher order; §??), the forced-fit null (the basis declines to manufacture a component where none is selected; §??), the asymmetry between the selected and forced populations (§??), the structurally independent shell decomposition flagging the same galaxies at approximately the same radii (§??–??), and the reproduction of the selection on independently derived rotation curves (§??) are all properties expected of galaxy-intrinsic structure and of none of an unrestricted flexible basis. The one place the curve-fitting concern does bite — the inner component's displacement below the halo scaling relation, which forcing reproduces — we therefore decline to interpret; the surviving scaling signal is the outer component's, not the inner's.

5. LIMITATIONS

Two limitations bound the claims. The concordant sample is modest (∼40 galaxies at the galaxy level; 28 at the radial level). The basis-independence and independent-derivation tests are partial: the former shares the data across methods, and the latter is currently established for the subset of galaxies whose curves can be re-derived from public cubes at sufficient quality, with one multi-domain system (NGC 1003) power-limited and one honest non-reproduction (NGC 3521) reported explicitly (§??). The stellar mass-to-light confound — the most direct baryonic route to a spurious second domain — is bounded directly by the free-Υ selection of §??, which leaves the fraction and the decisive population essentially intact while weakening individual margins; residual baryonic channels (radial Υ gradients, non-circular streaming from bars and spiral arms) are not excluded by it and are deferred, with the curve-quality characterization, to the larger-sample analysis in preparation.

A specific selection concern is that lower signal-to-noise curves might preferentially register as single-domain because a second domain is harder to resolve, inflating the single-domain fraction. Two features of the analysis bound this. The injection–recovery power controls of §?? quantify, per galaxy, the feature amplitude detectable given its sampling and uncertainties, and the non-reproductions there are classified as informative or power-limited on that basis rather than assumed. And the forced-fit null (§??) shows that added components in single-domain systems collapse to a degenerate morphology distinct from selected systems, which a signal-to-noise deficit alone would not produce. A full characterization of the multi-domain fraction as a function of curve quality is deferred to the larger-sample analysis in preparation. Confirmation against independent tracers of the same galaxies (in preparation) is the natural next test, and is also the route to the geometric discrimination that rotation curves cannot provide.

6. CONCLUSIONS

A minimum-order radial-domain decomposition of 121 SPARC rotation curves shows that ∼35% of galaxies require a second resolved radial mass domain beyond any single smooth equilibrium profile (NFW, Burkert, or Einasto), a result robust to confining all components within the measured radii — a constraint that simultaneously eliminates all three-domain solutions as artifacts. The detection is consistent across methods: an algorithmically distinct Gaussian-shell decomposition identifies the same galaxies (mass-adjusted odds ratio 37.5) and consistent radii (partial $\rho = 0.62$ controlling for the half-light radius), and when the domain model is fit first it absorbs the majority of those shells, identifying them as the same structure in a localized basis. Where it can be tested, the selection also survives independent velocity derivation: it reproduces on rotation curves re-derived from the original radio cubes by independent software, decisively and at the same radius. The component parameters do not organize on a privileged scale ratio or a common surface-density relation, and no galaxy property beyond mass and shell structure — including bulge presence — predicts the second domain. Where two domains are present, the single-profile scale radius is a biased compromise between the two domain scales (41/41 betweenness; median factor 2.2), so smooth fits both conceal the structure and distort the inferred core scale. The residual radial structure concealed by single smooth halo fits is therefore supported by convergent evidence, bounded, reproducible on independent data, and — as to its three-dimensional geometry and physical origin — deliberately left open, to be addressed with observables that carry the angular information rotation curves lack.

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